

# Miguel Angel Velásquez Vásquez

## Personal Information:

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## Education

- 31/12/2024    **PhD in Psychology**, Tulane University.  
*Dissertation:* Investigating the effects of musical training and neurofeedback on cognitive enhancement and brain plasticity. *Advisor:* Dr. Paul J. Colombo.
- 07/05/2021    **M.Sc. in Psychology**, Tulane University.  
*Thesis:* The effects of neural entrainment on sensorimotor synchronization. *Advisor:* Dr. Paul J. Colombo.
- 09/05/2014    **B.S. in Psychology with Honors**, University of New Orleans.  
*Thesis:* Reciprocal relation between psychophysiological patterns of stress responsivity and sleep. *Advisor:* Dr. Elizabeth A. Shirtcliff.

## Research Experience

- 2025 -  
Ongoing    **Postdoctoral Researcher** Department of Psychology, University of New Orleans.  
*Supervisor:* Dr. Debra Karhson
- Assisted with the data analysis and dissemination of research results including creation of grants, manuscripts, and presentations.
  - Collected EEG, saliva and blood analytes, and behavioral data from autistic and neurotypical participants.
  - Overseeing scheduling and tracking of participants, quality control of local data collection, & data management and data transfer.
  - Developed efficient data cleaning and analysis pipelines of periodic and aperiodic EEG data components, used as potential biomarkers for conscious states.
- 2024    **Researcher Assistant** Department of Economy, Tulane University.  
*Supervisor:* Dr. Patrick Button & Dr. Brigham Walker

- Conducted literature reviews and statistical analyses on factors contributing to race-related accessibility issues in mental health care.

2018-2024

**Researcher Assistant** Department of Psychology, Tulane University.

*Supervisor:* Dr. Paul Colombo

- Conducted data analysis and dissemination of research results including management of grant-funded projects, writing manuscripts, and presentations.
- Conducted a meta-analysis on the relationship between neurofeedback training parameters and neural modulation success.
- Performed ERP and behavioral analyses to determine the differences in conscious conflict monitoring and sensory processing between musicians and non-musicians.
- Collaborated on the experimental design, collection and analysis of fNIRS data to determine the impact of acute and chronic mindful meditation on hemodynamic brain activity.
- Collected and analyzed EEG and behavioral data on 60 participants to determine the prolonged effects of auditory stimuli on SSEP signals.
- Conducted cortisol and total protein extraction on saliva samples using ELISA method to test the effects of music training on stress reactivity.

2014

**Researcher Assistant** Department of Psychology, University of New Orleans.

*Supervisor:* Dr. Connie Lamm

- Operated as a Dense Array EEG Technician.
- Collected neurophysiological, and behavioral data from human participants.
- Performed data cleaning and entry on collected data.

2011 - 2014

**Researcher Assistant** Department of Psychology, University of New Orleans.

*Supervisor:* Dr. Elizabeth Shirtcliff

- Studied the effects of sleep quality on stress reactivity and salivary cortisol.
- Collected biological and behavioral data from human participants.
- Wet lab technician performing enzyme-immuno assays (ELISA) on salivary samples.

## Publications

(\*Denotes equal contribution to the work)

2025

Galang, E. J.\*, Velasquez, M. A.\*, Elcin, D., O'Connell, S., & Colombo, P. J. (2025). Systematic Review and Meta-Analysis of the Relationships between Real-Time Neurofeedback Training Parameters and Acquisition of Neural Modulation. *Frontiers in Human Neuroscience*, 19, 1652607.

2024

**Velasquez, M. A.,** Winston, J. L., Sur, S., Yurgil, K. A., Upman, A. E., Wroblewski, S. R., Huddle, A., & Colombo, P. J. (2024). Music training is related to late ERP modulation and enhanced performance during the Simon task but not the Stroop task. *Frontiers in Psychology*, 18, 1384179.

2024 Elcin, D., **Velasquez, M. A.,** & Colombo, P. J. (2024). Effects of acute and long-term mindfulness on the conflict resolution component of attention. *Frontiers in Human Neuroscience*, 18, 1359198.

2020 Yurgil, K. A., **Velasquez, M. A.,** Winston, J. L., Reichman, N. B., & Colombo, P. J. (2020). Music training, working memory, and neural oscillations: A review. *Frontiers in Psychology*, 11, 266.

## Works in Progress

2026 **Velasquez, M. A.,** Weinstein, C. R., Scott, M. V., & Karhson, D. (In Preparation). Auditory task-related aperiodic EEG activity in autism spectrum disorder (ASD).

2026 Brigham, W., **Velasquez, M. A.,** & Button, P. (In Preparation). Role of race concordance in mental health access.

## Participation at Professional Meetings

(\*Denotes collaborator-led presentations)

2026 "Auditory task-related aperiodic EEG activity in autism spectrum disorder (ASD)." *Poster presented at the Cognitive Neuroscience Society Conference*. Vancouver, Canada

2023 "Music training is related to late ERP modulation and enhanced performance during the Simon task but not the Stroop task." *Poster presented at Society for Neuroscience Conference*. Washington, D.C.

2023 \*\*"Effects of acute and long-term mindfulness on the conflict resolution component of attention." *Poster presented at Society for Neuroscience Conference*. Washington, D.C.

2023 "The influence of musicianship on the timing of cognitive conflict resolution in Stroop and Simon tasks." *Poster presented at Tulane Research, Innovation and Creativity Summit*. New Orleans, LA.

2023 \*\*"Effects of acute and long-term mindfulness on the conflict resolution component of attention." *Poster presented at Tulane Research, Innovation and Creativity Summit*. New Orleans, LA.

2022 "Rhythmic priming results in neural entrainment and persistent effects on sensorimotor synchronization." *Poster presented at Health Science Research Days, Tulane University*. New Orleans, LA.

2021 "Rhythmic priming results in neural entrainment and persistent effects on sensorimotor synchronization." *Poster presented at Society for Neuroscience Conference*. Chicago, IL.

2021 "Musical experience is related to the time course of changes in levels of the immune response marker interleukin-6 after exposure to acute social stress." *Poster presented at Neuromusic VII – Fondazione Mariani*. Aarhus, Denmark.

- 2015 \*"A neurobiological view on testosterone responsivity to reward and challenge." *Presented at the International Convention of Psychological Science (ICPS)*. Amsterdam, The Netherlands.
- 2014 "Reciprocal relation between psychophysiological patterns of stress responsivity and sleep." *Presented at Posters on the Hill*. Washington, D.C.
- 2014 "Reciprocal relation between psychophysiological patterns of stress responsivity and sleep." *Poster presented at Psychology Undergraduate Research Conference*. UCLA, Los Angeles, CA.

## Invited Talks

- 2026 "The effects of musical training and neurofeedback on cognitive enhancement and brain plasticity." Psychology Graduate Seminar Series, University of New Orleans.
- 2026 "Auditory task-related aperiodic EEG activity in autism spectrum disorder (ASD)" Cognitive Neuroscience Society Data Blitz, Vancouver.
- 2022 "The effects of neural entrainment on sensorimotor synchronization." The Robert E. Flowerree III Psychology Colloquium Series, Tulane University.

## Honors & Awards

- 2026 Marie Skłodowska-Curie Actions (MSCA) Seal of Excellence (94.4%).
- 2026 Post-Doctoral Fellow Award. Cognitive Neuroscience Society 2026
- 2014 Excellence in Psychology Award. 41st annual College of Sciences Honors Convocation
- 2013 S. Thomas Elder Award Most Promising Undergraduate Research. Annual College of Sciences Honors Convocation
- 2012 Supervised Undergraduate Research Experience (SURE). Experimental program to Stimulate Competitive Research (LAEPSCoR).

## Workshops

- 2026 (Upcoming) **Consciousness Commons.** Collaborative science workshop with open consciousness data (ASSC 2026 Satellite Workshop). Santiago, Chile.
- Selected participant in a competitive workshop for early-career researchers. Workshop includes opportunity to compete for \$25,000 in seed funding.
  - Collaborate in interdisciplinary teams to produce a preregistered study using open-access datasets.
- 2013 **Neurobiology at LUMCON.** Two-week Tulane University sponsored neuroscience workshop. Cocodrie, Louisiana.
- Studied cellular neurobiology through the application of different electrophysiological techniques including single-cell voltage clamping on rat hippocampal neurons and patch-clamping on Aplysia ganglion.

## Teaching: Mentorship

### Graduate Student Supervision

Charlie Weinstein, Mya Scott

### Honors Thesis Student Supervision

Charlotte Pearson

### Baccalaureate Student Supervision

Laura Domensch, Eldrick Galang, Conner Angelle, Ilyssa Muise, Abby Freedman, Savannah McNair, Liam Tierney

### AP Research Mentor

Eesha Dasai

## Skills

**Programming and Statistical Tools:** Matlab, R, Python, SPSS

**Neuroimaging Tools:** EEG, fNIRS, ERP

**Signal Processing:** Time-frequency analysis, power spectral density (PSD), dynamic and static functional connectivity, steady state evoked potentials (SSEP)

**Wet Lab Techniques:** Enzyme-linked immunosorbent assay (ELISA)

**Others:** E-Prime, PsychoPy, BCI

**Languages:** Spanish (Native), English (Fluent), French (Conversational)

## Service to the Field

### Ad Hoc Journal Reviewer

Frontiers in Psychology - Auditory Cognitive Neuroscience

Annals of the New York Academy of Sciences

Nature - Scientific Reports

Behavioral and Brain Functions Advances in Cognitive Psychology

## Membership in Professional Societies

Cognitive Neuroscience Society (CNS)

Society for Neuroscience (SfN)

American Psychological Association (APA)

Society for Music Perception and Cognition (SMPC)

Multidisciplinary Association for Psychedelic Studies (MAPS)

## Other

Ongoing

**Tutoring** Univariate Statistics and R Tutoring.

2023

**Siddhartha Opera** Interactive Music and Dance Performance, Music Co-composer

2023

**Walk-way Voices** Interactive Sound Art Installation, Greenway Supernova

- 2023 **First LEGO League Challenge Championship Judge**
- 2022 **Finalist in Novel Tech Challenge technology development program "Neurowaves: gamma wave neural stimulation through audiovisual stimulation"**